

The peritoneum is the largest serous membrane in the body, and its arrangements are complex. In males it forms a closed sac, but in females it is open at the lateral ends of the uterine tubes. Although its smooth appearance is unremarkable, its structure is complex and varies greatly in different locations. Directly beneath the monolayer of mesothelium is a well-developed basement membrane, outside which is a rich lymphatic plexus. Microscopic mesothelial pores or peritoneal stomata are distributed throughout the peritoneum but particularly on the undersurface of the diaphragm and anterior abdominal wall (Wassilev et al 1998). At peritoneal stomata, the mesothelium is in close proximity to underlying lymphatic endothelial cells and there is no intervening basement membrane. Both peritoneal stomata and sites of lymphoid aggregates ('milky spots') in the omenta contribute to peritoneal fluid absorption, and their occlusion by malignant cells may result in ascites (Hagiwara et al 1994). Neighbouring mesothelial cells are joined by junctional complexes but probably permit the passage of macrophages. The submesothelial connective tissue may also contain macrophages, lymphocytes and adipocytes (in some regions). Mesothelial cells may transform into fibroblasts, which may play an important role in inflammation of the peritoneum and the formation of peritoneal adhesions after surgery (Schnüriger et al 2011).

The normal pressure within the peritoneal cavity is about 2–10 mmHg (higher in pregnancy and obese individuals) (Sanchez et al 2001). It can be measured directly using a catheter inserted into the abdomen, or indirectly, by monitoring the pressure in the bladder or stomach. The pressure inside the abdomen may increase after trauma as a result of the accumulation of blood, fluid or oedema, leading to an 'abdominal compartment syndrome'. In such patients, monitoring of intra-abdominal pressure can identify major increases in pressure (above about 25 mmHg) that could jeopardize the blood flow to vital organs and dictate the need for urgent decompression.

PERITONEAL FLUID

The peritoneal cavity is a potential space between the parietal peritoneum, which lines the abdominal wall, and the visceral peritoneum, which covers the abdominal viscera and associated mesenteries within the cavity. It contains a small amount of peritoneal fluid that rarely exceeds 5 ml in healthy males and postmenopausal females (Yoshikawa et al 2013). In healthy young females, up to 25 ml of fluid may be present, depending on the phase of the menstrual cycle (Koninckx et al 1980). The fluid lubricates the mobile viscera, allowing them to glide freely on the abdominal wall and against each other within the limits imposed by their attachments. However, the parietal peritoneum can be surgically removed without necessarily having adverse effects on gut function (Sugarbaker 2012). Normal peritoneal fluid contains water, proteins (less than 30 g of protein per litre), electrolytes and solutes derived from interstitial fluid in the adjacent tissues and from plasma in local capillaries. It also contains a few cells, including desquamated mesothelium, nomadic peritoneal macrophages, mast cells, fibroblasts, lymphocytes and other leukocytes. Macrophages migrate freely between the peritoneal cavity and the surrounding connective tissue. In females, blood or fluid may enter the peritoneal cavity from a ruptured ovarian follicle (mittelschmerz) or from the retrograde flow of menstrual fluid along the uterine tubes (which may cause endometriosis). Peritoneal fluid is absorbed via peritoneal stomata and milky spots (see above). Under normal circumstances, the peritoneal cavity never contains gas.

The peritoneum and its fluid have important defensive properties (Autio 1964). Peritoneal fluid gravitates to dependent sites within the peritoneal cavity; diaphragmatic respiratory movements, negative intrathoracic pressure and intestinal peristalsis encourage flow from the pelvis to the subphrenic regions, even when the individual is erect. Fluid preferentially ascends from the lower abdomen and pelvis to the right subhepatic and subphrenic 'spaces' via the right paracolic gutter, which is deeper than the left paracolic gutter and has no obstructing phrenico-

colic ligament (Meyers 1973). Passage across the midline to the left subphrenic space is prevented by the falciform ligament. Peritoneal fluid flows in a predominantly clockwise direction around the peritoneal cavity (Fig. 63.1); it reaches the surface of the greater omentum, where it is processed by the immune system (Meyers et al 1987, Meyers 1994). Malignant cells may become trapped within peritoneal recesses or in the milky spots of the greater omentum, where they may proliferate and produce visible and palpable tumour deposits. These sites must be deliberately examined when searching for peritoneal metastases (Zonca et al 2008). Gross infiltration of the omentum by cancer, often referred to as an 'omental cake', emphasizes the capacity of the omentum to trap malignant cells. The movement of peritoneal fluid explains the presence of disease at sites such as the undersurface of the right hemidiaphragm and the pelvic pouches. An example is the Fitz-Hugh-Curtis syndrome, in which gonorrhoea or chlamydia organisms enter the peritoneal cavity through the uterine tubes and are transported in the peritoneal fluid to the right upper quadrant, where they cause perihepatic inflammation. Disruption of the integrity of the peritoneum by percutaneous puncture, laparoscopy or open surgery profoundly affects its defensive function and can create 'sticky' sites where pathogens or malignant cells can settle and proliferate.

Direct communication of abdominal and pleural spaces called pleuroperitoneal fenestrae may rarely exist (Pestieu et al 2000, Simmons and Mir 1989).

PERITONEAL ATTACHMENTS

The parietal peritoneum is attached to the muscular layers of the abdominal wall by extraperitoneal connective tissue. Around the bladder and rectum, it is only loosely attached to allow for alterations in the size of these viscera, whereas it is firmly adherent to the inferior surface of the diaphragm and linea alba. The extraperitoneal tissue over the posterior abdominal wall frequently contains large amounts of fat, especially in obese males. The visceral peritoneum is firmly adherent to the underlying viscera and often blends with connective tissue in the wall of the organ; it rarely contains loose connective or adipose tissue. The visceral peritoneum is often considered as part of the underlying viscus for clinical and pathological purposes such as the staging of cancer. Because of its attachments, removal of the parietal peritoneum in the management of peritoneal metastases is possible without resection of underlying tissues, whereas cancer deposits on visceral peritoneum usually require partial resection of the viscus for complete removal (see Video 63.1).

GENERAL ARRANGEMENT OF THE PERITONEUM

The alimentary tract develops as a single tube suspended in the coelomic cavity by ventral and dorsal mesenteries (Ch. 60). Ultimately, the ventral mesentery is largely resorbed, although some parts persist in the upper abdomen and form structures such as the lesser omentum and falciform ligament. The mesenteries of the intestines in the adult are the remnants of the dorsal mesentery. The migration and subsequent fixation of parts of the gastrointestinal tract produce the so-called 'retroperitoneal' segments of bowel (most of the duodenum, ascending colon, descending colon, and rectum) (Fig. 63.2), and four separate intraperitoneal bowel loops suspended by mesenteries of variable lengths. These are all covered by visceral peritoneum, which is continuous with the parietal peritoneum covering the posterior abdominal wall. The first intraperitoneal loop is formed by the abdominal oesophagus, stomach and first part of the duodenum. The second loop is made up of the duodenojejunal junction, jejunum, ileum and usually the caecum. The third loop contains the transverse colon, and the final loop contains the sigmoid colon and occasionally the distal descending colon.

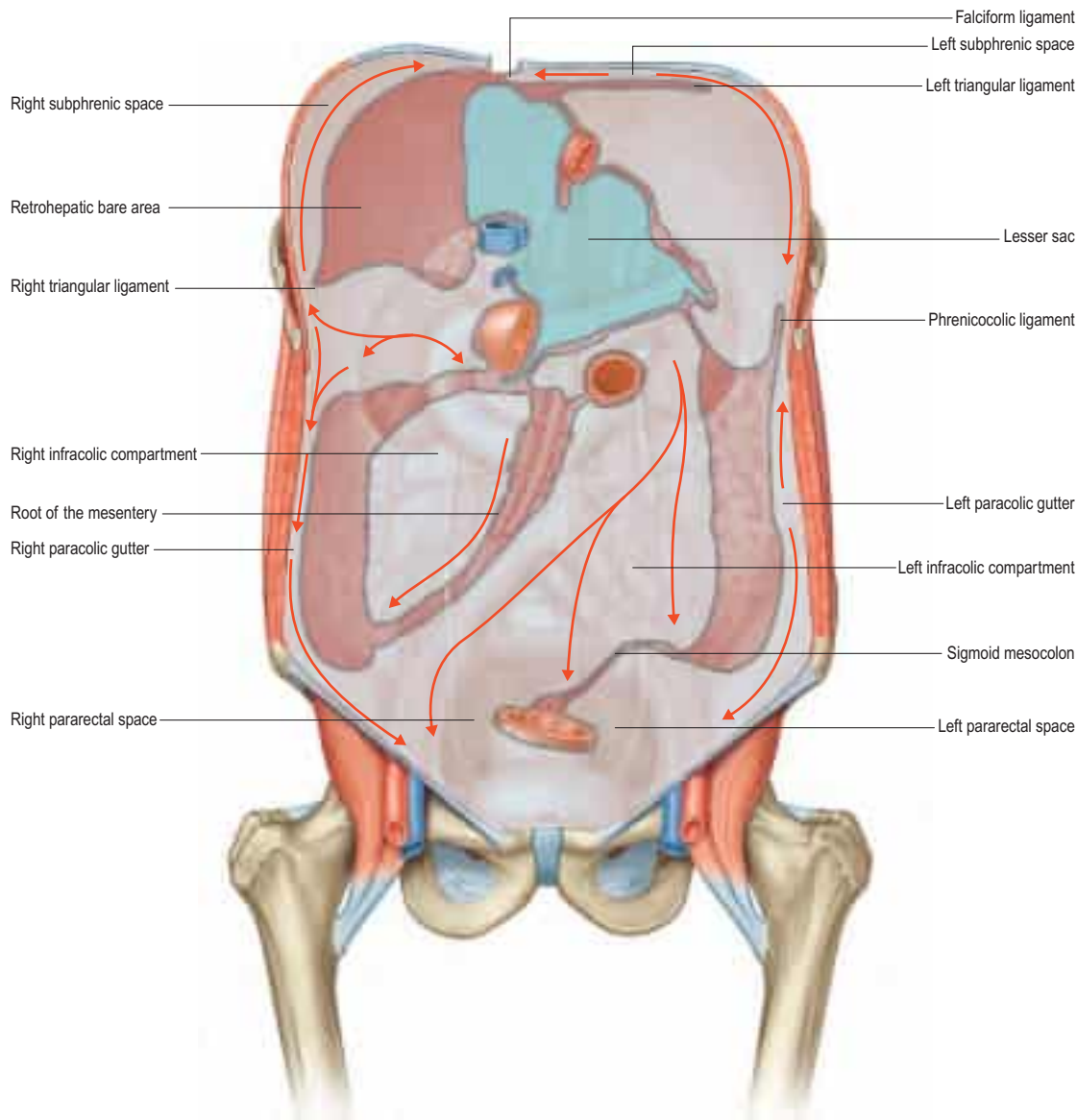


Fig. 63.1 The peritoneal cavity. Arrows indicate the flow of peritoneal fluid. (Courtesy of CM Brown, Office of Visual Media IUSM 2011.)

Where visceral peritoneum encloses or suspends organs within the peritoneal cavity, the peritoneum and related connective tissues form 'peritoneal ligaments', omenta or mesenteries. All but the greater omentum are composed of two layers of visceral peritoneum separated by variable amounts of fatty connective tissue. The greater omentum is folded back on itself and therefore consists of four layers of visceral peritoneum separated by variable amounts of adipose tissue. Mesenteries attach their respective viscera to the posterior abdominal wall; the attachment of the mesentery of the small intestine to the posterior abdominal wall is referred to as the root of the mesentery (see Fig. 63.2). The mesenteries contain the neurovascular bundles and lymphatic channels that supply the suspended organs. In obese individuals, extensive adipose tissue within the mesenteries and omenta may obscure these neurovascular bundles. In contrast, in the very young, the elderly or the malnourished, the mesenteries and omenta may contain little adipose tissue and the neurovascular bundles are more obvious.

Although they are described as intraperitoneal, the suspended viscera, strictly speaking, do not lie within the peritoneal cavity because they are covered by visceral peritoneum. They are in continuity with extraperitoneal tissues, including the retroperitoneum. The loose areolar connective tissue that forms the extraperitoneal and retroperitoneal tissues is often divided into potential 'spaces' as evidenced by the presence of discrete collections of fluid or blood in pathological conditions (see Figs 62.4–62.6).

PERITONEUM OF THE UPPER ABDOMEN

The abdominal oesophagus, stomach, liver and spleen all lie within a double fold of visceral peritoneum that runs from the posterior to the anterior abdominal wall. This fold has no recognized name but has been referred to as the mesogastrium, from which it is derived (Coakley

and Hricak 1999). It has a complex attachment to the wall of the abdominal cavity and forms the falciform ligament, coronary ligaments, lesser omentum (gastrohepatic and hepatoduodenal ligaments), greater omentum (including the gastrocolic ligament), gastrosplenic ligament, splenorenal ligament and phrenicocolic ligament (see Fig. 63.2; Figs 63.3–63.5).

Falciform ligament

The falciform ligament is a thin anteroposterior double fold of peritoneum that connects the liver to the posterior aspect of the anterior abdominal wall just to the right of the midline. Adjacent to the anterior abdominal wall, it contains a variable amount of fat (Feldberg and van Leeuwen 1990). It extends inferiorly to the level of the umbilicus and superiorly it narrows to a depth of 1–2 cm as the distance between the liver and anterior abdominal wall decreases. Its two peritoneal layers separate to enclose the liver, to which it is firmly adherent. Superiorly, the two peritoneal layers are continuous with the parietal peritoneum on the undersurface of the diaphragm but are reflected laterally to form the superior layer of the coronary ligament of the liver on the right and the left triangular ligament of the liver on the left. The inferior aspect of the falciform ligament forms a free border, where the two peritoneal layers are continuous with each other as they enclose the ligamentum teres. Because the peritoneum of the falciform ligament is in continuity with that covering the posterior abdominal wall and with the parietal peritoneum of the anterior abdominal wall, blood from a retroperitoneal haemorrhage (commonly, acute haemorrhagic pancreatitis) may track between the folds of peritoneum and appear as haemorrhagic discoloration around the umbilicus (Cullen's sign). The bloodstained fluid reaches the periumbilical region via the lesser omentum and falciform ligament or via the pararenal spaces and abdominal wall.

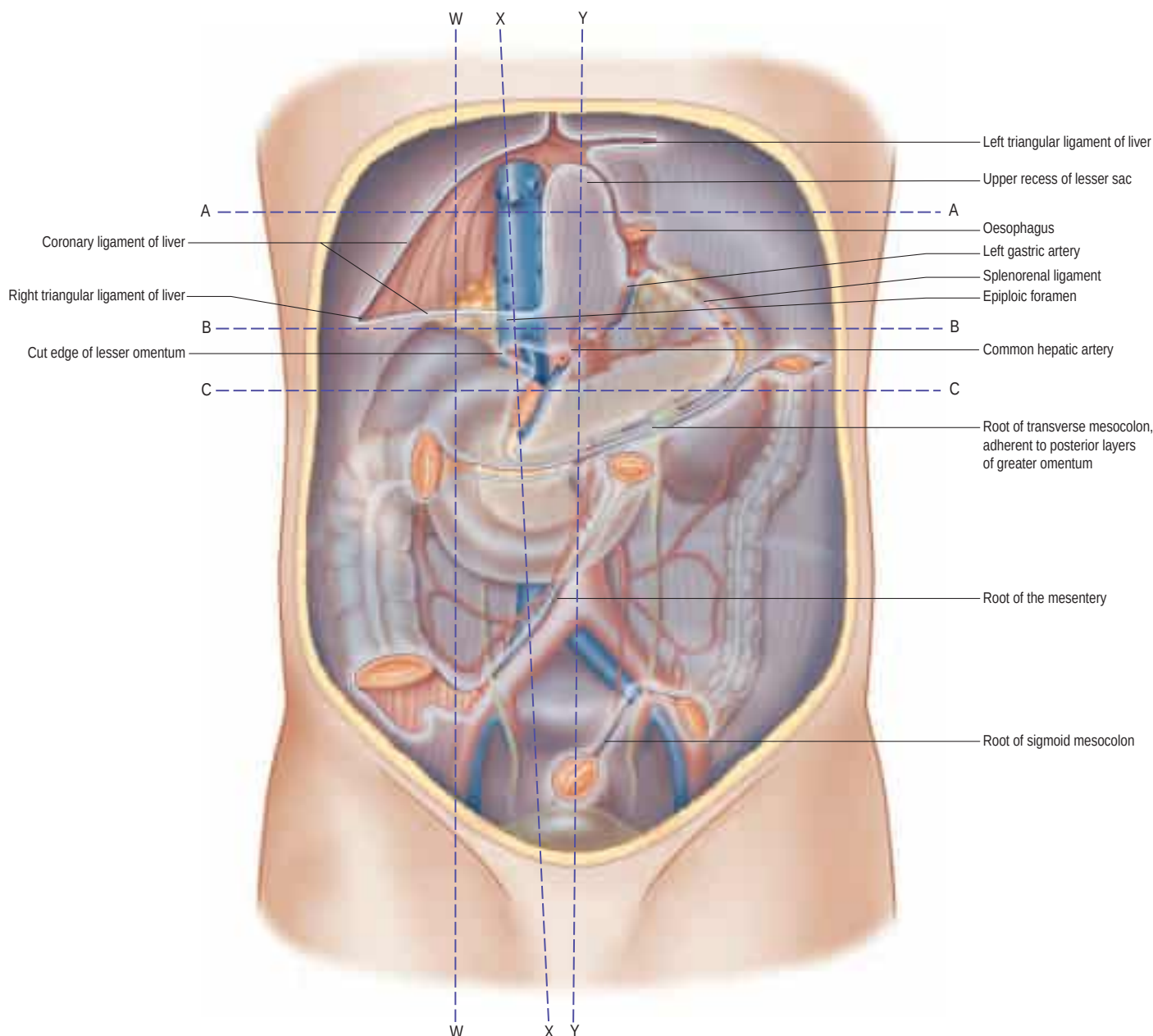


Fig. 63.2 The posterior abdominal wall, showing the lines of peritoneal reflection, after removal of the liver, spleen, stomach, jejunum, ileum, caecum, transverse colon and sigmoid colon. The various sessile (retroperitoneal) organs can be seen through the posterior parietal peritoneum. Note the ascending and descending colon, duodenum, kidneys, suprarenal glands, pancreas and inferior vena cava. Line W-W represents the plane of [Figure 63.3](#). Line X-X represents the plane of [Figure 63.4](#). Line Y-Y represents the plane of [Figure 63.5](#). Line A-A represents the plane of [Figure 63.6A](#). Line B-B represents the plane of [Figure 63.6B](#). Line C-C represents the plane of [Figure 63.6C](#).

Peritoneal connections of the liver

The liver is almost completely covered in visceral peritoneum; only the 'bare area' posteriorly is in direct contact with the right hemidiaphragm. Peritoneal ligaments run from the liver to the surrounding viscera and to the abdominal wall and diaphragm (see [Fig. 63.2](#); [Fig. 63.6](#)); they are described in detail in [Chapter 67](#).

The coronary ligament is formed by the reflection of the peritoneum from the diaphragm on to the superior and posterior surfaces of the right lobe of the liver. Between the two layers of this ligament, a large triangular area, the 'bare area' of the liver, is devoid of peritoneal covering. Here, the liver is connected to the diaphragm by areolar tissue, which is in continuity inferiorly with the anterior pararenal space. On the right, the two layers of the coronary ligament converge laterally to form the right triangular ligament. On the left, the two layers fuse to form the left triangular ligament. The upper layer of the coronary ligament is reflected superiorly on to the undersurface of the diaphragm and inferiorly on to the right and superior surfaces of the liver. The lower layer of the coronary ligament is reflected inferiorly from the posterior surface of the liver on to the posterior abdominal wall over the right suprarenal gland and kidney. The peritoneal recess formed between the inferior surface of the liver and the upper pole of the right

kidney is known as the hepatorenal pouch (of Morison) ([Stringer 2009](#)). In the supine position, this is the most dependent part of the peritoneal cavity in the upper abdomen and is a site where fluid or peritoneal metastases may localize.

The lower layer of the coronary ligament is continuous with the peritoneum that descends over the anterior surface of the right kidney and to the front of the first part of the duodenum and hepatic flexure of the colon (see [Fig. 63.2](#)). Medially, it passes in front of a short segment of the inferior vena cava lying between the first part of the duodenum below and the caudate process of the liver above, forming the posterior wall of the epiploic foramen. This narrow strip broadens out as the peritoneum continues across the midline on to the posterior wall of the lesser sac.

The right triangular ligament is a short, V-shaped fold at the right lateral limit of the bare area of the liver, where the two layers of the coronary ligament converge. It is continuous with the peritoneum of the right posterolateral abdominal wall. Surgical division of the right triangular and coronary ligaments allows the right lobe of the liver to be retracted forwards, exposing the lateral aspect of the inferior vena cava behind the liver.

The left triangular ligament is a double layer of peritoneum that extends for a variable length over the superior border of the left lobe

of the liver. Medially, the anterior leaf is continuous with the left layer of the falciform ligament, while the posterior layer is continuous with the lesser omentum. The left triangular ligament lies in front of the abdominal oesophagus, the upper limits of the lesser omentum, and part of the fundus of the stomach. Division of the left triangular liga-

ment allows the left lobe of the liver to be mobilized. The left triangular ligament is an important stabilizing factor for the left lobe after excision of the right lobe of the liver.

The peritoneum of the left triangular ligament is continuous with that lining the undersurface of the left dome of the diaphragm, which, in turn, is continuous with that lining the posterior abdominal wall. Here it passes behind the spleen on to the most lateral part of the mesentery of the transverse colon and the splenic flexure. It continues down lateral to the descending colon into the pelvis, forming the left paracolic gutter. Medially, the peritoneum covering the left upper posterior abdominal wall is reflected anteriorly to merge with the upper end of the lesser omentum, the peritoneum overlying the left aspect of the abdominal oesophagus, and the left layer of the gastrosplenic ligament (see Fig. 63.2).

Parietal peritoneum lines the posterior abdominal wall over the diaphragmatic crura, the upper abdominal aorta and its associated branches, lymph nodes and nerve plexuses, and the upper anterior surface of the pancreas. Below the liver, it continues down the posterior abdominal wall, forming the right paracolic gutter between the antero-lateral abdominal wall and the ascending colon.

Lesser omentum

The lesser omentum is formed by two layers of peritoneum separated by a variable amount of connective tissue and fat. It is derived from the ventral mesogastrium. It runs between the inferior visceral surface of the liver and the abdominal oesophagus, stomach, pylorus and first part of the duodenum. Superiorly, its attachment to the inferior surface of the liver forms an L shape. The vertical component of the L is formed by the fissure for the ligamentum venosum; inferiorly, the attachment turns horizontally to complete the L in the porta hepatis. The part of the lesser omentum running between the liver and the stomach is known as the gastrohepatic ligament and the part between the liver and duodenum is the hepatoduodenal ligament. The right and left gastric vessels, branches of the vagus nerves and lymph nodes are contained within the two layers of the gastrohepatic ligament close to its gastric attachment. An accessory or replaced left hepatic artery is sometimes present (p. 1166). The anterior layer of the lesser omentum descends from the fissure for the ligamentum venosum on to the anterior surface of the abdominal oesophagus, stomach and duodenum. The posterior layer descends from the fissure and runs on to the posterior surface of the stomach and pylorus. At the lesser curvature of the stomach, the peritoneal layers of the lesser omentum split to enclose the stomach

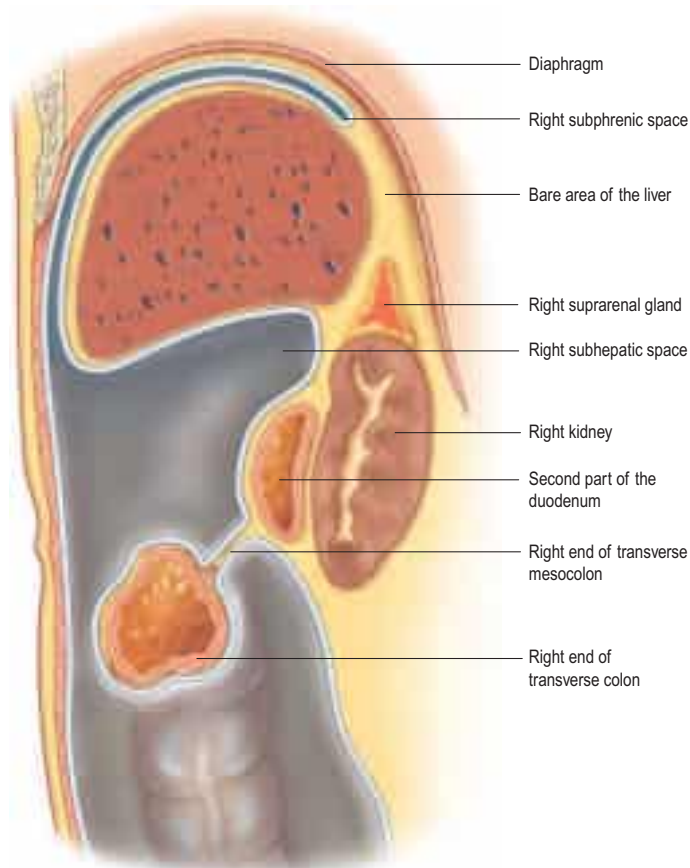


Fig. 63.3 A sagittal section through the abdomen to the right of the epiploic foramen along the line of W-W in Figure 63.2.

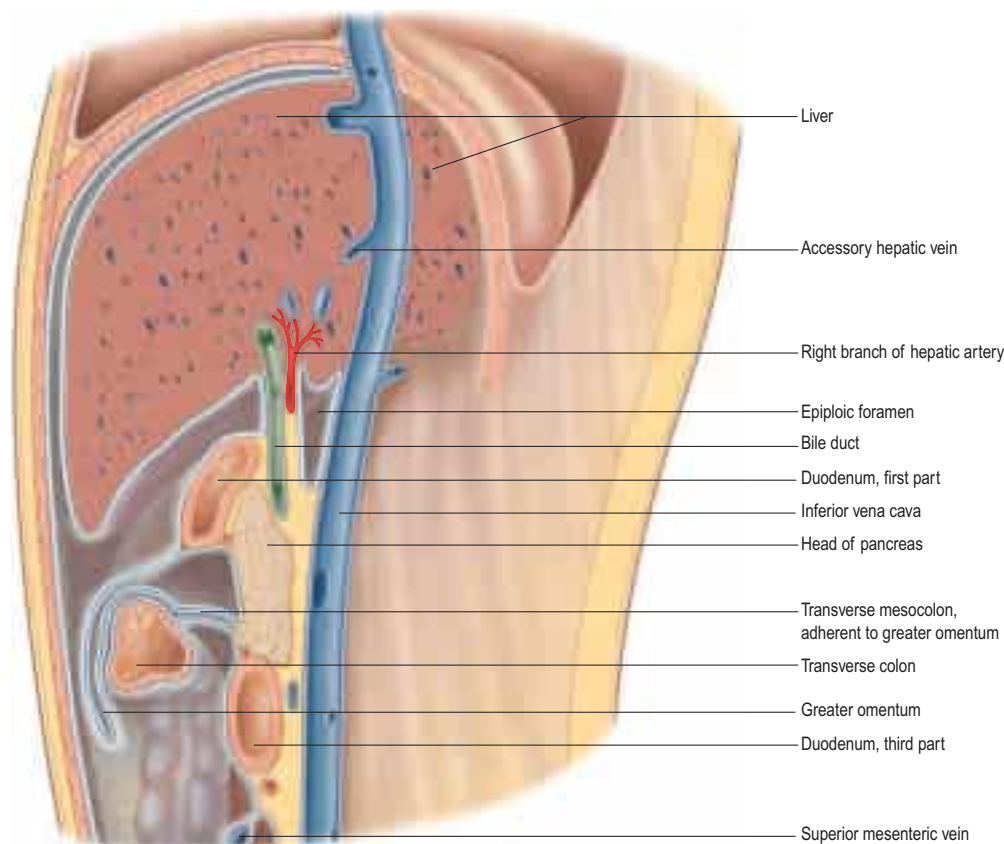


Fig. 63.4 A section through the upper part of the abdominal cavity, along the line X-X in Figure 63.2. The boundaries of the epiploic foramen are shown and a small recess of the lesser sac is displayed anterior to the head of the pancreas. Note that the transverse colon and its mesocolon are adherent to the posterior two layers of the greater omentum. Four peritoneal layers contribute to the formation of the greater omentum because the apron of omentum is folded back on itself.

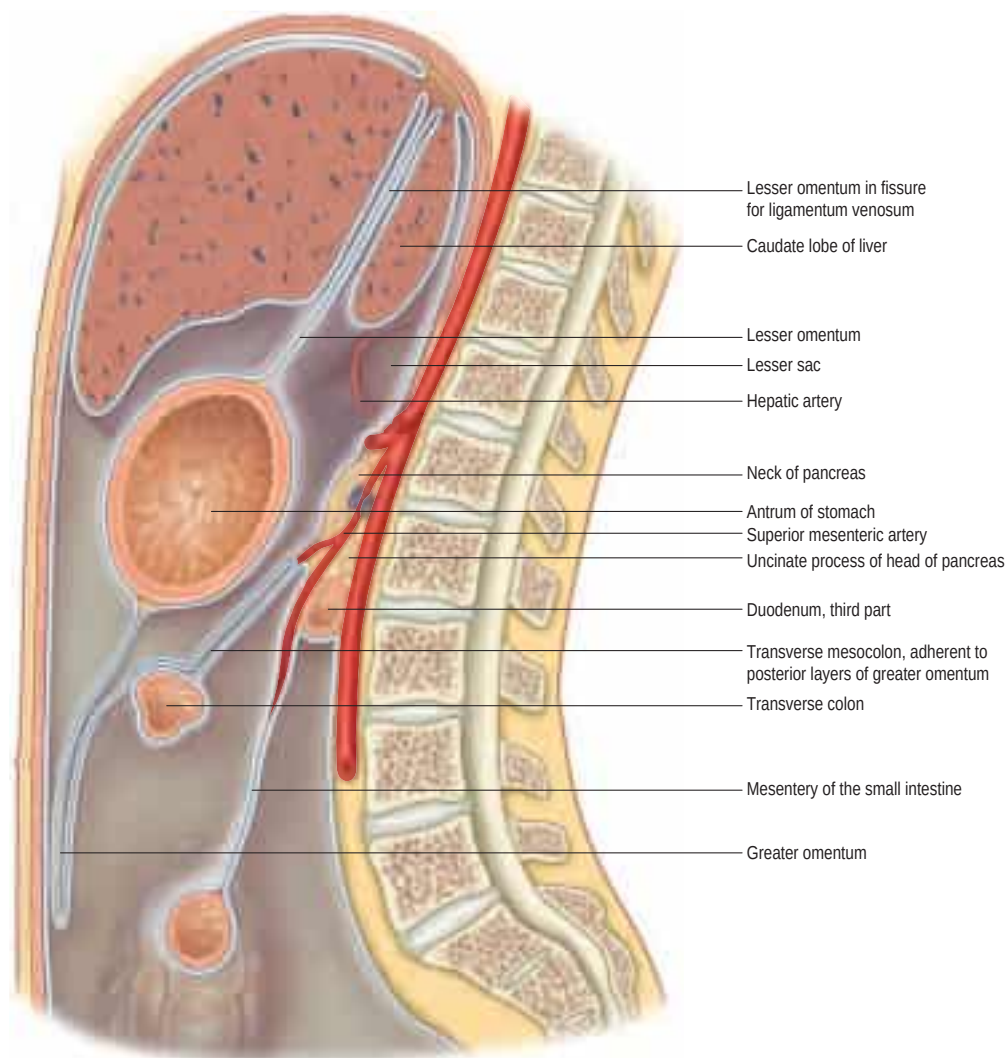


Fig. 63.5 A sagittal section through the abdomen, approximately in the median plane. Compare with [Figure 63.2](#). The section cuts the posterior abdominal wall along the line Y–Y in [Figure 63.2](#). The peritoneum is shown in blue except along its cut edges, which are left white.

and are continuous with the visceral peritoneum covering the anterior and posterior surfaces of the stomach. The posterior layer of the lesser omentum forms part of the anterior surface of the lesser sac (see [Fig. 63.5](#)).

The free right lateral border of the lesser omentum is thickened and extends from the porta hepatis to the junction between the first and second parts of the duodenum. This free border forms the anterior boundary of the epiploic foramen and contains the portal vein (posteriorly), the bile duct (anteriorly to the right), and hepatic artery proper (anteriorly to the left) (see [Fig. 63.6B](#)), together with lymphatics, lymph nodes and hepatic autonomic nerves. An accessory or replaced right hepatic artery arising from the superior mesenteric artery may sometimes be found in the free edge of the lesser omentum, usually running posterior to the portal vein. Occasionally, the free margin of the lesser omentum extends to the right and runs on to the gallbladder, when it is referred to as the cystoduodenal ligament ([Ashaolu et al 2011](#)).

The upper border of the lesser omentum is short and runs over the inferior surface of the diaphragm between the liver and medial aspect of the abdominal oesophagus. The lesser omentum is thinner in this region and may be fenestrated or incomplete; variations in thickness are dependent on the amount of connective tissue, especially fat.

Greater omentum

The greater omentum is the largest peritoneal fold and hangs inferiorly from the greater curvature of the stomach. When the undisturbed abdomen is opened, the greater omentum is frequently draped over the upper abdominal organs, but often not in the even distribution shown in many illustrations. It is usually thin but always contains some adipose tissue and is a common site for storage of fat in obese individuals, particularly males ([Coulrier 2009](#)). The greater omentum is a double sheet: each sheet consists of two layers of peritoneum separated by a scant amount of connective tissue. The two sheets are folded back on themselves and firmly adherent to each other below the transverse colon. Above this level the anterior sheet separates from the posterior

sheet and is attached to the greater curve of the stomach (see [Fig. 63.5](#)); this supracolic part of the greater omentum is known as the gastrocolic ligament. The most anterior layer of peritoneum of the greater omentum is continuous with the visceral peritoneum over the anterior surface of the stomach and duodenum. The anterior sheet descends from the greater curvature of the stomach and first part of the duodenum for a variable distance into the peritoneal cavity and then turns sharply backwards on itself to ascend as the posterior sheet. The posterior sheet passes anterior to the transverse colon and transverse mesocolon, and is attached to the posterior abdominal wall above the origin of the transverse mesocolon and anterior to the head and body of the pancreas (see [Figs 63.2, 63.5](#)). The anterior peritoneal layer of the posterior sheet is continuous with the peritoneum of the posterior wall of the lesser sac. The posterior peritoneal layer is reflected sharply inferiorly and is continuous with the anterior layer of the transverse mesocolon. The posterior sheet is adherent to the transverse colon and its mesentery. In early fetal life, the greater omentum and transverse mesocolon are separate structures, and this arrangement occasionally persists. During surgery, the natural adhesions between the transverse mesocolon and greater omentum can be divided, and the greater omentum can be detached from the transverse colon and mesocolon if required. Access into the lesser sac can then be obtained by dividing the upper part of the posterior sheet of greater omentum. This gives a relatively bloodless surgical approach to the posterior wall of the stomach and the anterior surface of the pancreas.

On the left side of the abdomen, the greater omentum is continuous with the gastrosplenic ligament; on the right side, it extends to the beginning of the duodenum (see [Fig. 63.2](#)). A fold of peritoneum, the hepatocolic ligament, may run from either the inferior surface of the right lobe of the liver or the first part of the duodenum to the right side of the greater omentum or hepatic flexure of the colon. The right border of the greater omentum is usually adherent to the anterior surface of the hepatic flexure and upper ascending colon but its peritoneal layers are not continuous with the visceral peritoneum over this part of the colon. The right border of the greater omentum may also be attached to a thin but vascular peritoneal band (Jackson's membrane),

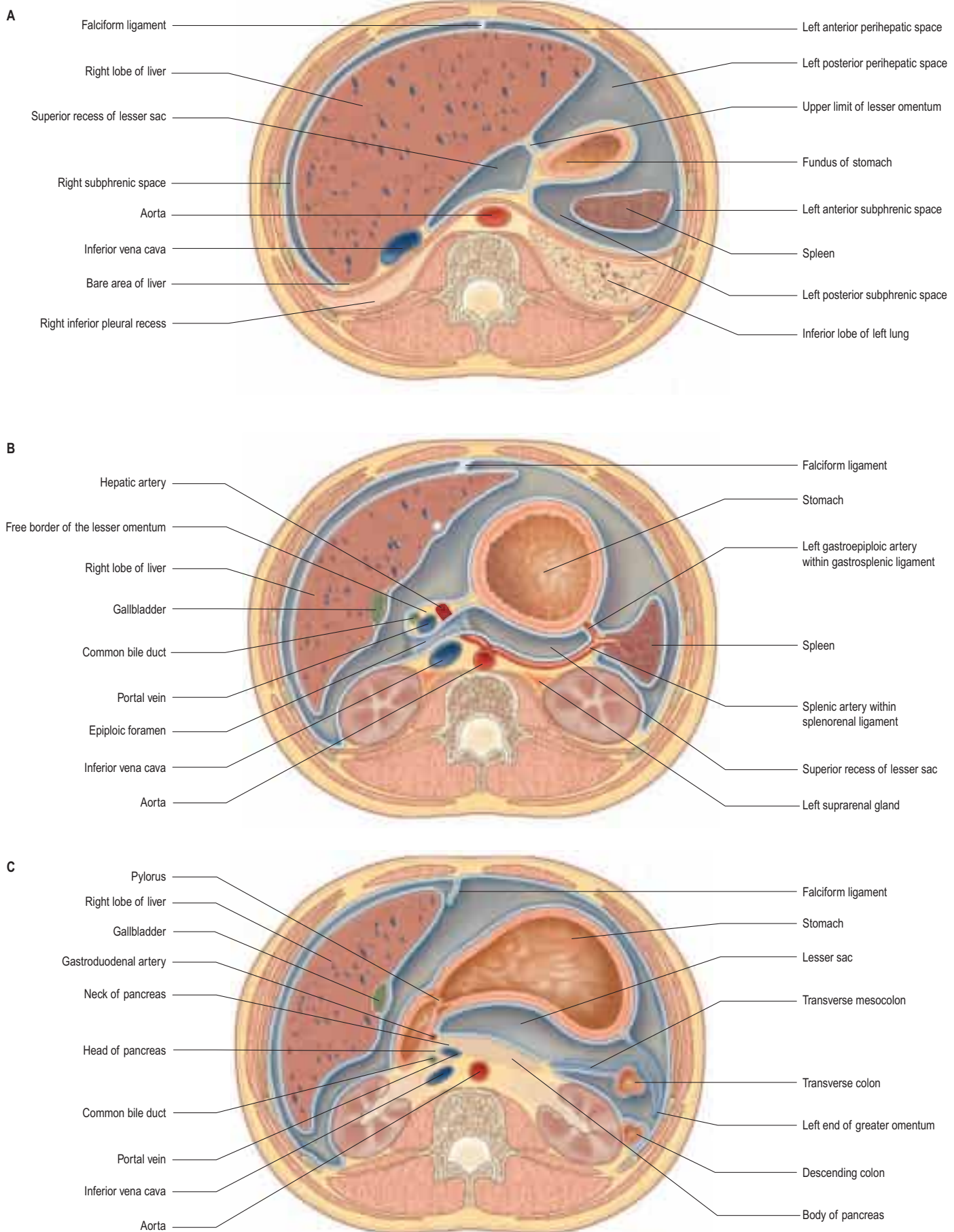


Fig. 63.6 Transverse sections through the abdomen. **A**, At the level of the line A–A in Figure 63.2. The line passes through the bare area of the liver at the superior end of the lesser omentum. The parts of the left subphrenic space are clearly seen, although they are continuous with each other. **B**, At the level of line B–B in Figure 63.2, viewed from below. The peritoneal cavity is shown by blue shading; the peritoneum and its cut edges are in pale blue. **C**, A transverse section through the abdomen at the level of the line C–C in Figure 63.2, viewed from below. Colours as in Figure 63.2.

which, when present, runs from the anterior aspect of the ascending colon and caecum to the posterolateral abdominal wall. Other peritoneal folds between the ascending colon and posterolateral abdominal wall in the right lateral paracolic gutter are occasionally present. Less commonly, the left border of the greater omentum is adherent to the anterior surface of the descending colon.

The greater omentum has a rich blood supply derived from the right and left gastropiploic vessels that run between the two peritoneal layers of the anterior sheet of greater omentum, close to the greater curvature of the stomach. These vessels give off numerous omental (epiploic) branches that run the length of the omentum. Microscopic capillary convolutions, sometimes termed 'omental glomeruli', lie directly beneath the mesothelium of the greater omentum. They are surrounded by aggregations of macrophages and lymphocytes (milky spots) that drain into lymphatic channels (Shimotsuma et al 1993); the macrophages form clusters near the mesothelial surface, while B and T lymphocytes are grouped around vessels. The fenestrated endothelium lining the omental glomeruli and the mesothelium overlying the milky spots share a similar mesodermal origin and together they form sites of leukocyte migration and peritoneal fluid absorption (Platell et al 2000). The number and size of milky spots increase in the presence of peritoneal inflammation, highlighting their defensive role.

The greater omentum serves several important functions. It is highly mobile and frequently becomes adherent to inflamed viscera or foreign bodies within the abdominal cavity. This can limit the spread of infection, as in the walling off of acute appendicitis, and promote haemostasis. However, it may also block peritoneal drains and catheters (such that omentectomy is sometimes performed when placing some intraperitoneal catheters). It absorbs peritoneal fluid. It contains macrophages and lymphoid tissue capable of destroying pathogens; macrophages and neutrophils readily migrate from the greater omentum into the peritoneal cavity in the presence of peritonitis. It promotes neovascularization of structures to which it becomes adherent and thus promotes healing of ischaemic tissues and supports splenic autotransplantation (Ch. 70). It is rarely the site of primary pathology; reports of segmental infarction, torsion or other disorders are rare.

The remarkable capacity of the greater omentum to participate in reparative processes is highlighted by its use as an omental pedicle flap or free flap in reconstructive surgery. It has been used to close gastrointestinal perforations, fill 'dead space' after surgical excision of pathology, and cover wound defects. A greater omental pedicle or free flap based on the right gastropiploic vessels may be mobilized by dividing the gastric branches of these vessels close to the surface of the stomach and the left gastropiploic vessels near their origin.

Peritoneal connections of the spleen

The peritoneal connections of the spleen anchor it in the left upper quadrant of the abdomen; they include the gastrosplenic, splenorenal and phrenicocolic ligaments. The gastrosplenic ligament runs between the greater curvature of the stomach and the hilum of the spleen, and is in continuity with the left side of the greater omentum. The peritoneal layers of the gastrosplenic ligament separate to enclose the spleen and then rejoin to form the splenorenal and phrenicocolic ligaments. The splenorenal ligament extends from the spleen to the posterior abdominal wall, and the phrenicocolic ligament connects the splenic flexure of the colon to the diaphragm.

The splenorenal ligament is formed from two layers of peritoneum (see Fig. 63.6B). The anterior layer is continuous medially with the peritoneum of the posterior wall of the lesser sac over the left kidney and runs up to the splenic hilum, where it is continuous with the posterior layer of the gastrosplenic ligament. The posterior layer of the splenorenal ligament is continuous laterally with the peritoneum over the inferior surface of the diaphragm and runs on to the splenic surface over the renal impression. The splenic vessels lie between the layers of the splenorenal ligament, and the tail of the pancreas is usually present in its lower portion.

The gastrosplenic ligament is also formed from two layers of peritoneum (see Fig. 63.6B). The posterior layer is continuous with the peritoneum of the splenic hilum and the peritoneum over the posterior surface of the stomach. The anterior layer is continuous with the peritoneum reflected off the gastric impression of the spleen and with the peritoneum over the anterior surface of the stomach. The short gastric and left gastropiploic branches of the splenic artery, with their corresponding veins, pass between the layers of the gastrosplenic ligament. A fan-shaped fold of peritoneum often extends from the anterior aspect of the gastrosplenic ligament below the inferior pole of the spleen and blends with the phrenicocolic ligament. The phrenicocolic ligament

extends from the splenic flexure of the colon to the diaphragm at about the level of the eleventh rib. Inferiorly, it is continuous with the peritoneum of the lateral end of the transverse mesocolon near the tip of the pancreatic tail, and the splenorenal ligament at the hilum of the spleen.

Undue traction on the peritoneal attachments of the spleen at surgery may cause a capsular tear and serious bleeding. Downward traction on the phrenicocolic ligament during mobilization of the splenic flexure is particularly hazardous; medial traction on the greater omentum may also injure the spleen if the omentum is adherent to the splenic capsule.

PERITONEUM OF THE LOWER ABDOMEN

The posterior surface of the lower anterior abdominal wall is lined by parietal peritoneum, which extends from the linea alba centrally to the lateral paracolic gutters, where it is reflected over the front and sides of the ascending colon on the right and the descending colon on the left (see Fig. 63.2). Occasionally, the ascending and descending colon are suspended by a short mesentery from the posterior abdominal wall (Ch. 66). Between the ascending and descending colon, the peritoneum covers the posterior abdominal wall, except where it is reflected over the oblique root of the mesentery of the small intestine. Over the posterior abdominal wall, it covers psoas major and quadratus lumborum, the inferior vena cava, duodenum, vertebral column and both ureters. In the upper abdomen, the posterior parietal peritoneum is reflected anteriorly where it is continuous with the peritoneum of the posterior layer of the transverse mesocolon.

Transverse mesocolon

The mesentery of the transverse colon is a broad fold of visceral peritoneum reflected anteriorly from the posterior abdominal wall. It suspends the transverse colon in the peritoneal cavity. The root of the transverse mesocolon passes obliquely from the anterior aspect of the second part of the duodenum, over the head and neck of the pancreas, superior to the duodenojejunal junction over the upper pole of the left kidney to the splenic flexure (see Fig. 63.2). It varies considerably in length but is shortest at either end. It contains the middle colic vessels and their branches, together with autonomic nerves from the aortic plexus, lymphatics and lymph nodes. Near the splenic flexure, the ascending branch of the left colic artery terminates within the transverse mesocolon. Its two peritoneal layers pass to the posterior surface of the transverse colon, where they separate to invest the colon. The upper layer of peritoneum is continuous with the posterior layer of the greater omentum, to which it is adherent (see Fig. 63.5). The lower layer of peritoneum of the transverse mesocolon is continuous with the peritoneum of the posterior abdominal wall. Lateral extensions of the transverse mesocolon produce two folds: on the right, the duodenocolic ligament extends from the transverse mesocolon at the hepatic flexure to the second part of the duodenum; on the left, the phrenicocolic ligament extends from the transverse mesocolon at the splenic flexure to the diaphragm. The root of the transverse mesocolon is closely related to the upper limit of the root of the mesentery of the small intestine near the uncinate process of the pancreas.

Mesentery of the small intestine

The mesentery of the small intestine is arranged as a complex fan that is formed from two layers of peritoneum (anterosuperior and postero-inferior) separated by fatty connective tissue containing vessels and nerves. The root of the mesentery lies along a line running diagonally from the duodenojejunal flexure on the left side of the second lumbar vertebral body to the right sacroiliac joint (see Fig. 63.2). The root crosses over the third part of the duodenum, abdominal aorta, inferior vena cava, right ureter and right psoas major. The average length of the root of the mesentery is 15 cm in adults while along its intestinal attachment it measures the same length as the small intestine (approximately 5 m). Consequently, the mesentery is usually thrown into multiple folds along its intestinal border. The average depth of the mesentery from the root to the intestinal border is 20 cm, but this varies along the length of the small intestine: it is shortest at the jejunum and terminal ileum, and longest in the mid-ileal region. Its two peritoneal layers contain the jejunal and ileal branches of the superior mesenteric vessels, nerves related to the superior mesenteric plexus, lymphatics and regional lymph nodes. The length and mobility of the mesentery may make it difficult to determine the site and orientation of a loop of small intestine through a small surgical incision. Tracing the continuity of a

peritoneal layer of the mesentery on to the posterior abdominal wall may be useful in orientating an isolated loop if the duodenojejunal flexure or ileocaecal junction is not readily accessible. The mesentery of the small intestine is sometimes joined to the transverse mesocolon at the duodenojejunal flexure by a peritoneal band. Occasionally, the fourth part of the duodenum possesses a very short mesentery that is continuous with the upper end of the root of the mesentery of the small intestine. Additional bands of peritoneum may connect the terminal ileum to the posterior abdominal wall. The root of the mesentery of the small intestine is continuous with the peritoneum that surrounds the appendix and caecum in the right iliac fossa.

Mesoappendix

The mesentery of the appendix is a fatty, triangular fold of peritoneum that passes between the posterior surface of the mesentery of the terminal ileum close to the ileocaecal junction and the vermiform appendix. It may end short of the tip of the appendix, in which case a thin, shallow fold of peritoneum containing fat is present towards the tip. It encloses the blood vessels, nerves and lymph vessels of the vermiform appendix, and usually contains a lymph node. A small fold of peritoneum runs between the terminal ileum and the anterior layer of the mesoappendix (the 'bloodless fold of Treves'), and another fold of peritoneum containing the anterior caecal artery may extend from the terminal ileal mesentery to the anterior wall of the caecum (see below).

Sigmoid mesocolon

The sigmoid mesocolon varies in length and depth between individuals. The root of the sigmoid colon usually has a shallow, inverted V-shaped attachment; the apex of the V overlies the division of the left common iliac artery (see Fig. 63.2). Alternatively, the attachment may form a gentle curve. The left limb of the attachment runs over the left psoas major and along the external iliac vessels. The right limb passes over the pelvic brim towards the midline at the level of the third sacral vertebra. The anteromedial peritoneal layer of the sigmoid mesocolon is continuous with the peritoneum of the lower left posterior abdominal wall, and its posterolateral layer is continuous with the peritoneum of the lateral abdominal wall. Bands of peritoneum may be present running from the proximal sigmoid colon to the posterior abdominal wall. The sigmoid and superior rectal vessels run between its layers and the left ureter descends into the pelvis behind its apex and anterior to the bifurcation of the left common iliac artery.

Peritoneum of the lower anterior abdominal wall

The peritoneum of the lower anterior abdominal wall is raised into five folds (sometimes referred to as 'ligaments'), which diverge as they descend from the umbilicus. They are the median, right and left medial, and right and left lateral, umbilical folds (Fig. 63.7, see Fig. 75.9). The median umbilical fold extends from the umbilicus to the apex of the bladder and contains the urachus or its remnant (Ch. 72). The medial umbilical fold overlies the obliterated umbilical artery, which ascends from the internal iliac artery in the pelvis to the umbilicus. The lateral umbilical fold covers the inferior epigastric vessels below their entry into the rectus sheath. The suprapavesical fossa lies between the medial and median umbilical folds on each side of the midline, and the medial and lateral inguinal fossae lie on either side of each lateral umbilical fold. The lateral inguinal fossa overlies the deep inguinal ring, and the medial inguinal fossa overlies the femoral ring (Healy and Reznick 1999).

PERITONEUM OF THE PELVIS

The parietal peritoneum of the abdominal wall is continuous with the parietal peritoneum that lines the side walls of the pelvis. The arrangement of the pelvic visceral peritoneum differs between the sexes.

Peritoneum of the male pelvis

In males, the peritoneum of the left lower abdominal wall is reflected from the distal sigmoid colon and anterolateral surface of the upper rectum to the side walls of the true pelvis, where it forms the right and left pararectal fossae (Fig. 63.8); these vary in size according to the degree of distension of the rectum. The peritoneum is reflected anteri-

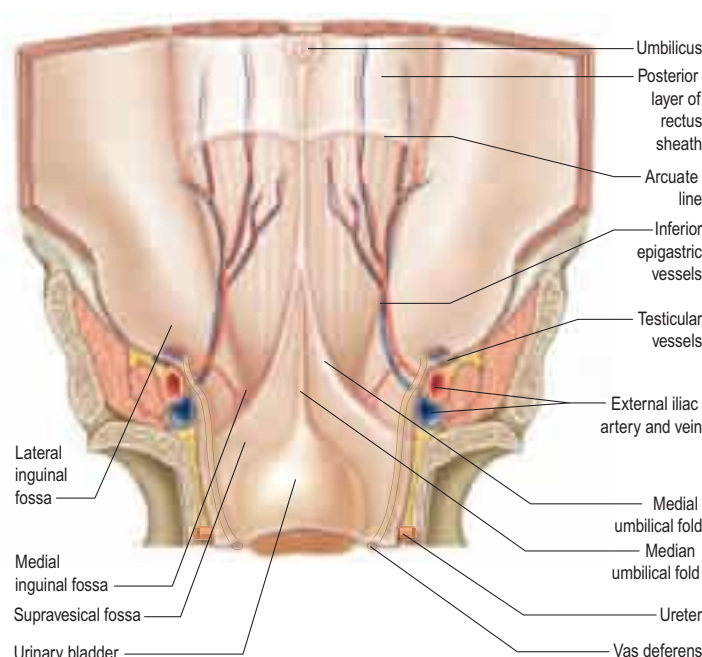


Fig. 63.7 The infra-umbilical part of the anterior abdominal wall of a male subject: posterior surface, with the peritoneum *in situ*.

only from the anterior surface of the rectum over the upper poles of the seminal vesicles and on to the posterior surface of the bladder, producing the rectovesical pouch. The pouch is limited laterally by sacrogenital folds of peritoneum, which extend posteriorly from the sides of the bladder to the anterior aspect of the sacrum. The peritoneum covering the posterosuperior surface of the bladder forms a paravesical fossa on each side that is limited laterally by a ridge of peritoneum containing the vas deferens; the depth of the paravesical fossae depends on the volume of urine in the bladder. When the bladder is empty, a transverse vesical fold may be visible at laparoscopy; its medial portion overlies the superior vesical artery or arteries (Boaz et al 2011). Between the paravesical and pararectal fossae, the ureters and internal iliac vessels may cause slight elevations in the peritoneum. From the apex of the bladder, the median umbilical fold extends superiorly on the posterior surface of the lower anterior abdominal wall to the umbilicus. When the bladder distends, the overlying peritoneum is lifted so that part of the anterior surface of the bladder comes into direct contact with the posterior surface of the lower anterior abdominal wall. This means that a distended bladder can be punctured directly through the lower anterior abdominal wall without traversing the peritoneal cavity (suprapubic puncture).

Peritoneum of the female pelvis

In females, peritoneum covers the anterolateral surface of the upper rectum as it does in the male, but it descends further over the anterior surface of the rectum. The pararectal and paravesical fossae are limited laterally by the peritoneum that covers the uterosacral and round ligaments of the uterus, respectively (see Figs 77.14, 77.15, 77.18). The presence of the uterus and vagina means that there are two pelvic pouches instead of the single rectovesical pouch seen in males. The peritoneum from the rectum is reflected anteriorly on to the posterior surface of the posterior fornix of the vagina and the uterus, forming the recto-uterine pouch (of Douglas). The depth of the pouch – namely, the extent to which it descends on the posterior surface of the vagina – is variable (Baessler and Schuessler 2000). The peritoneum covers the fundus and body of the uterus, descending on its anterior surface as far as the junction of the body and cervix; from here, it is reflected forwards on to the posterosuperior surface of the bladder, forming a shallow vesico-uterine pouch. As in males, the peritoneum over the dome of the bladder is reflected on to the posterior surface of the lower anterior abdominal wall. Recto-uterine folds containing the uterosacral ligaments pass posteriorly from the sides of the cervix to the sacrum, running lateral to the rectum. Peritoneum is also reflected from the anterior and posterior uterine surfaces to the lateral pelvic walls as the broad ligament of the uterus (see Fig. 77.18). This consists of anterior and posterior layers that are continuous at the upper border of the ligament. The broad ligament contains the uterine tubes in its free upper border; the ovaries are suspended from its posterior layer. Inferiorly, the

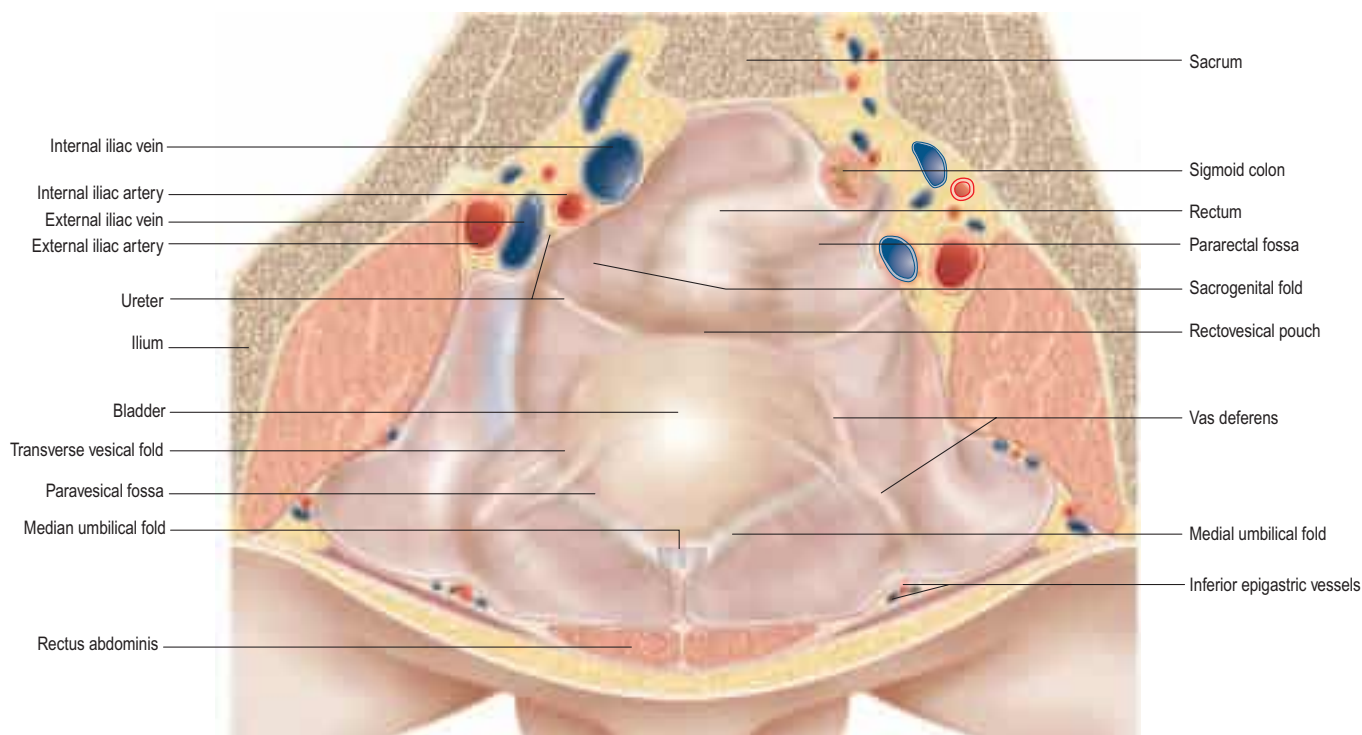


Fig. 63.8 The peritoneum of the male pelvis: anterosuperior view.

peritoneum at the base of the broad ligament merges with the pelvic parietal peritoneum. The ovarian fossa is a shallow depression in the peritoneal lining of the lateral pelvic wall between the peritoneal ridges formed by the obliterated umbilical arteries anteriorly, the ureter posteriorly, and the external iliac vessels above; it lies behind the lateral attachment of the broad ligament and usually contains the ovary in nulliparous females.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Parietal and visceral peritoneum develops from the somatopleural and splanchnopleural layers, respectively, of the lateral plate mesoderm (p. 1059). Parietal peritoneum is supplied by the somatic blood vessels of the abdominal and pelvic walls, and its lymphatics join those in the body wall and drain to parietal lymph nodes. Visceral peritoneum is best considered as an integral part of the viscus it covers: its blood supply and lymphatic drainage therefore correspond to those of the associated viscus.

INNERVATION

The parietal peritoneum is innervated by somatic efferent and afferent nerves that also supply the muscles and skin of the overlying body wall. The visceral peritoneum is innervated by afferent nerves that travel with the autonomic supply to the underlying viscera. Sensations arising from pathologies that affect the parietal or visceral peritoneum reflect these different patterns of innervation. Well-localized pain is elicited by mechanical, thermal or chemical stimulation of nociceptors in the parietal peritoneum; the pain is usually well localized to the affected region. Somatic nerves that innervate the parietal peritoneum also supply the corresponding segmental skin and muscles; when the parietal peritoneum is irritated, local reflex muscle contraction occurs, resulting in clinical signs of guarding or even rigidity of the abdominal wall. The parietal diaphragmatic peritoneum is supplied centrally by afferent fibres from the phrenic nerves and peripherally by the lower intercostal and subcostal nerves; peripheral irritation of the diaphragm may therefore result in pain localized in the distribution of the lower thoracic spinal nerves, while central irritation causes referred pain in the cutaneous distribution of the third to fifth cervical spinal nerves (the shoulder region). The innervation of the parietal peritoneum of the true pelvis is poorly documented but the obturator nerve makes a significant contribution.

Irritation or stretch of the visceral peritoneum causes poorly localized discomfort. The sensation of pain is referred to a diffuse area of the abdominal wall. Thus, discomfort from structures derived from

the foregut is felt in the epigastric region, that from midgut structures in the umbilical region, and that from hindgut structures in the suprapubic region; none of these sensations is significantly lateralized. Pain from the pancreas, gallbladder and even the small bowel may radiate to the back. Stretch or irritation of the visceral peritoneum may also elicit profound reflex vasomotor and cardiac changes mediated by autonomic nerves, including a vasovagal response. This is of considerable clinical relevance. Painful responses to manipulation of the parietal peritoneum may be abolished by local or regional anaesthesia. In marked contrast, the direct central connections of visceral afferents, particularly via the vagus nerve, mean that stretching the visceral peritoneum may induce profound effects, including acute haemodynamic instability, despite local or regional (including spinal) anaesthesia.

GENERAL ARRANGEMENT OF THE PERITONEAL CAVITY

The peritoneal cavity is a single continuous space between the parietal peritoneum lining the abdominal wall and the visceral peritoneum enveloping the abdominal organs. It consists of the greater sac, which is the main peritoneal cavity surrounding most of the abdominal and pelvic viscera, and the lesser sac, or omental bursa, which is a small diverticulum situated behind the stomach and lesser omentum and in front of the pancreas. These two sacs communicate via the epiploic foramen.

For clinical purposes, the peritoneal cavity can be divided into several 'spaces'; pathological processes are often contained within these spaces and their anatomy may influence diagnosis and treatment. It is useful to divide the peritoneal cavity into two main compartments, supramesocolic (often simply called supracolic) and inframesocolic (or infracolic), which are partially separated by the transverse colon and its mesentery. The pelvic peritoneal spaces are described above.

SUPRAMESOCOLIC COMPARTMENT

This lies between the diaphragm and the transverse mesocolon. It can be arbitrarily divided into right and left supramesocolic spaces. The right supramesocolic space can be subdivided into the right subphrenic space, the right subhepatic space and the lesser sac. The left supramesocolic space can be divided into two subspaces: the left subphrenic space and the left perihepatic space. These 'spaces' usually communicate but may nevertheless be sites of localized fluid collections.

Right subphrenic space

The right subphrenic space lies between the diaphragm and the anterior, superior and right lateral surfaces of the right lobe of the liver. It is bounded on the left by the falciform ligament and posteriorly by the upper layer of the coronary ligament (see Fig. 63.3). Because of the clockwise flow of peritoneal fluid from the lower abdomen and pelvis, it is a relatively common site for an infected fluid collection after a perforated appendicitis or duodenal ulcer.

Right subhepatic space (hepatorenal pouch)

The right subhepatic space lies between the inferior surface of the right lobe of the liver and the upper pole of the right kidney. It is bounded superiorly by the inferior layer of the coronary ligament, laterally by the right lateral abdominal wall, posteriorly by the anterior surface of the upper pole of the right kidney, and inferomedially by the hepatic flexure, transverse mesocolon, second part of the duodenum, and part of the head of the pancreas. In the supine position, the hepatorenal pouch (of Morison) is more dependent than the right paracolic gutter. It is a site where a pathological fluid collection may develop.

Lesser sac (omental bursa)

The lesser sac is a cavity lined with peritoneum and connected to the main peritoneal cavity (greater sac) by the epiploic foramen (of Winslow). It is considered part of the right supramesocolic compartment because it develops in the embryo on the right side of the ventral mesogastrium (p. 1059). It has posterior and anterior walls and superior, inferior, right and left borders. The sac varies in size according to the volume of the viscera making up its walls; it may be partially obliterated by natural adhesions between the anterior and posterior walls.

The anterior wall of the lesser sac consists of the posterior peritoneal layer of the lesser omentum, the peritoneum over the posterior wall of the stomach and first part of the duodenum, and the posterior upper part of the anterior sheet of the greater omentum (see Fig. 63.5). At its right border, the anterior wall is mostly formed by the lesser omentum but, towards the left, the lesser omentum becomes progressively shorter and more of the anterior wall is formed by the posterior surface of the stomach and greater omentum.

The lower part of the posterior wall of the lesser sac is formed by the anterior peritoneal layer of the posterior sheet of the greater omentum, which overlies the transverse mesocolon. More superiorly, the peritoneum of the posterior wall covers, from below upwards, a small part of the head and the whole neck and body of the pancreas, the medial part of the anterior aspect of the upper pole of the left kidney, most of the left suprarenal gland, the commencement of the abdominal aorta, the coeliac trunk and part of the diaphragm (see Fig. 63.2). The inferior phrenic, splenic, left gastric and common hepatic arteries lie partly behind the bursa. Many of these structures form the 'bed' of the stomach and are separated from it only by the linings of the lesser sac.

The superior border of the lesser sac is narrow and lies between the right side of the oesophagus and the upper end of the fissure for the ligamentum venosum; it is sometimes referred to as the superior recess of the omental bursa. Here, the peritoneum of the posterior wall of the lesser sac is reflected anteriorly from the diaphragm to join the posterior layer of the lesser omentum. The inferior border runs along the line of fusion of the layers of the greater omentum, which extends from the gastrosplenic ligament on the left to the peritoneal fold behind the first part of the duodenum on the right. If the layers of the greater omentum are not completely fused, the lesser sac may extend to the inferior limit of this structure, but this is rarely demonstrable at surgery.

The right border of the lesser sac is where the peritoneum overlying the head and neck of the pancreas is reflected on to the inferior aspect of the first part of the duodenum. The line of this reflection ascends to the left, along the medial side of the gastroduodenal artery. Near the upper duodenal border, the right border joins the floor of the epiploic foramen round the hepatic artery proper. The epiploic foramen therefore interrupts the right border. Above the epiploic foramen, the right border is formed by the peritoneal reflection from the diaphragm to the right margin of the caudate lobe of the liver, which then crosses the inferior vena cava. The left border of the lesser sac runs from the left end of the root of the transverse mesocolon and is mostly formed by the inner peritoneal layers of the splenorenal and gastrosplenic ligaments (see Fig. 63.6). The part of the lesser sac lying between the splenorenal and gastrosplenic ligaments is referred to as the splenic recess. Above the spleen, the two ligaments merge to form the short gastrophrenic ligament, which passes forwards from the diaphragm to

the posterior aspect of the fundus of the stomach and forms part of the upper left border of the lesser sac. The two layers of the gastrophrenic ligament diverge near the abdominal oesophagus, leaving part of the posterior gastric surface devoid of peritoneum. The left gastric artery runs forwards here into the lesser omentum.

The lesser sac is indented by a crescentic peritoneal fold that runs from the upper border of the neck of the pancreas to the upper part of the lesser curvature of the stomach. It is variably described but commonly known as the gastropancreatic fold. The upper part of this fold overlies the left gastric artery as it runs from the posterior abdominal wall to the lesser curvature of the stomach. The lower part of the fold overlies the common hepatic artery as it runs from the posterior abdominal wall to the lesser omentum. When the fold is prominent, it divides the lesser sac into a smaller superior and a larger inferior recess. The superior recess lies posterior to the lesser omentum and encloses the caudate lobe of the liver; it extends superiorly into the fissure for the ligamentum venosum and lies adjacent to the right crus of the diaphragm posteriorly. The inferior recess of the lesser sac lies between the stomach and pancreas, and is contained within the double sheet of the greater omentum.

Acute pancreatitis is probably the most common cause of a fluid collection within the lesser sac (or just behind its posterior peritoneal lining) (p. 1182). Bleeding from trauma or a ruptured splenic artery aneurysm and perforation of a posterior gastric ulcer are other causes of lesser sac collections.

Epiploic foramen

The epiploic foramen (foramen of Winslow or aditus to the lesser sac) is a short, vertical slit, about 3 cm in height in adults, behind the free right border of the lesser omentum. It is the entrance to the lesser sac from the greater sac. The hepatoduodenal ligament, formed by the thickened free right margin of the lesser omentum and extending between the porta hepatis above and the upper border of the first part of the duodenum below, forms the anterior boundary of the foramen. Within this free border lie the bile duct (anteriorly on the right), the hepatic artery (anteriorly on the left) and the portal vein (posteriorly), together with nerves and lymphatics (see above). Rapid control of the hepatic artery and portal vein can be obtained by compression of the free edge of the lesser omentum (a 'Pringle' manoeuvre), which is a potentially useful technique in liver trauma and surgery. Superiorly, the peritoneum of the posterior layer of the hepatoduodenal ligament runs over the caudate process of the liver, forming the roof of the epiploic foramen. This layer of peritoneum is then reflected on to the inferior vena cava, which forms the posterior border of the foramen. The floor of the foramen is formed by the peritoneal reflection overlying the upper border of the first part of the duodenum as it runs forwards above the head of the pancreas. A narrow passage, the vestibule of the lesser sac, may be formed to the left of the foramen between the caudate process above and the first part of the duodenum below. To the right, the rim of the foramen is continuous with the peritoneum of the greater sac. The anterior and posterior walls of the foramen are normally apposed, which partly explains why patients can develop large fluid collections isolated to the greater or lesser sac (Shrestha et al 2010).

Left subphrenic space

The left subphrenic space lies between the diaphragm, the anterior and superior surfaces of the left lobe of the liver, the anterosuperior surface of the stomach and the diaphragmatic surface of the spleen. To the right it is bounded by the falciform ligament. It is expanded in the absence of the spleen and is a common site for fluid collection after splenectomy. The left subphrenic space is substantially larger than the right and can be divided into a left perihepatic space and anterior and posterior subphrenic spaces (see Fig. 63.6); these are in continuity in the absence of disease. The definitions of the boundaries of these spaces vary.

The left anterior subphrenic space is large and lies between the superior and anterolateral surfaces of the spleen and the diaphragm. Inferiorly and medially, this space is bounded by the phrenicocolic, splenorenal and gastrosplenic ligaments. The phrenicocolic ligament partially obstructs the flow of fluid from the left paracolic gutter (Meyers 1973), which may explain why left subphrenic collections are less frequent than right subphrenic collections after lower abdominal and pelvic surgery. Nevertheless, the left subphrenic space is a relatively common site of fluid collection after upper abdominal surgery, particularly surgery involving the spleen or distal pancreas. The left posterior subphrenic space is small and lies between the fundus of the stomach and the diaphragm above the origin of the splenorenal ligament.

The left perihepatic space is sometimes subdivided into anterior and posterior spaces. The left anterior perihepatic space lies between the



Free tumour cells in the peritoneal cavity, especially from mucinous tumours, may gain access to the lesser sac and settle by gravity into the inferior recess of the lesser sac. This is more likely on the right side of the lesser sac in an area recognized surgically as the 'subpyloric space'. Growth of tumour at this site may cause gastric outlet obstruction.

During surgical exploration of the abdomen, some peritoneal surfaces cannot be visualized without the aid of specific surgical manoeuvres. One of these sites is the posterior aspect of the hepatoduodenal ligament. The recess created by the posterior aspect of the hepatoduodenal ligament and the caudate process is an important site for mucinous peritoneal metastases. It can be visualized during cytoreductive surgery by passing a tape around the structures in the free edge of the lesser omentum and retracting them forwards.

PERITONEUM AND PERITONEAL CAVITY

anterosuperior surface of the left lobe of the liver and diaphragm. The left posterior perihepatic space, also known as the gastrohepatic recess, lies inferior to the left lobe of the liver. It extends into the fissure for the ligamentum venosum on the right. Posteriorly, it is separated from the superior recess of the lesser sac by the lesser omentum.

Extraperitoneal subphrenic spaces

There are two potential subphrenic spaces that actually lie outside the peritoneum but are of clinical relevance because they may be sites of fluid accumulation. The right extraperitoneal space is bounded by the two layers of the coronary ligament, the bare area of the liver, and the inferior surface of the right hemidiaphragm. The left extraperitoneal space lies anterior to the left suprarenal gland and upper pole of the left kidney.

INFRAESOCOLIC COMPARTMENT

The inframesocolic compartment (also called the infracolic compartment) lies below the transverse mesocolon and transverse colon, and is divided into two unequal spaces by the root of the mesentery of the small intestine (see Fig. 63.2). It contains the right and left paracolic gutters lateral to the ascending and descending colon, respectively. As a consequence of the mobility of the transverse mesocolon and mesentery of the small intestine, and the open boundaries of the inframesocolic compartment, disease processes are rarely confined to the compartment. Continuous peristalsis of the small bowel may explain the relative lack of peritoneal metastases on intestinal and mesenteric surfaces.

Right infracolic space

The right infracolic space is a triangular space. It is smaller than its left-sided counterpart and lies posterior and inferior to the transverse colon and mesocolon to the right of the small intestinal mesentery. The space is narrowest inferiorly because the attachment of the root of the mesentery of the small intestine lies well to the right of the midline.

Left infracolic space

The left infracolic space is larger than its right-sided counterpart and is in free communication with the pelvis. It lies posterior and inferior to the transverse colon and mesocolon, and to the left of the mesentery of the small intestine. The sigmoid colon and its mesentery may partially restrict the flow of fluid or blood from this space into the pelvis to the left of the midline.

Paracolic gutters

The right and left paracolic gutters are peritoneal depressions on the posterior abdominal wall alongside the ascending and descending colon, respectively. The principal paracolic gutter lies lateral to the colon on each side. A less obvious medial paracolic gutter may be present, more often on the right side, if the ascending or descending colon possesses a short mesentery for part of its length. The right (lateral) paracolic gutter runs from the superolateral aspect of the hepatic flexure of the colon, down the lateral aspect of the ascending colon and caecum. It is continuous with the peritoneum of the pelvic cavity below. Superiorly, it is continuous with the peritoneum that lines the hepatorenal pouch and with the lesser sac through the epiploic foramen. Bile, pus, blood or other fluid may run along the gutter and collect in sites distant to the organ of origin. In supine patients, infected fluid from the right iliac fossa may ascend in the gutter to the right subphrenic space. In erect or semi-recumbent positions, fluid from the stomach, duodenum or gallbladder may run down the gutter to collect in the right iliac fossa (mimicking acute appendicitis) or pelvis to form an abscess. The right paracolic gutter is deeper than the left, which, together with the partial barrier provided by the phrenicocolic ligament, may explain why subphrenic collections are more common on the right.

RECESSES OF THE PERITONEAL CAVITY

Peritoneal fossae or recesses within the peritoneal cavity are occasionally sites of internal herniation. If a loop of intestine becomes stuck in

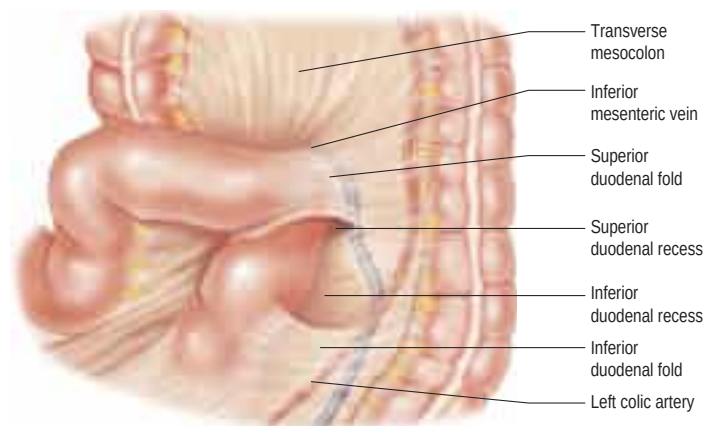


Fig. 63.9 The superior and inferior duodenal recesses. Note that the transverse colon and jejunum have been displaced.

a fossa or recess, the bowel may become obstructed or strangulate from a constriction at the entrance to the recess. The contents of the peritoneal fold forming the fossa/recess must be considered when repairing such a hernia.

Lesser sac

Although not usually classified as a peritoneal recess, the lesser sac may be a site of internal herniation. There are reports of the small intestine, caecum, transverse colon or gallbladder migrating through the epiploic foramen into the lesser sac and causing acute abdominal symptoms (Shrestha et al 2010).

Duodenal recesses

Several folds of peritoneum may exist around the fourth part of the duodenum and the duodenojejunal junction, forming a number of named recesses (Fig. 63.9). They probably arise during development as a result of minor aberrations of duodenal rotation and fixation. Some are potential sites of internal herniation (Zonca et al 2008).

Superior duodenal recess

The superior duodenal recess is occasionally present, often in association with an inferior duodenal recess. It lies directly to the left of the fourth part of the duodenum, adjacent to the second lumbar vertebra, and behind a crescentic superior duodenal fold. The fold has a semilunar free lower margin, which merges laterally with the peritoneum overlying the left kidney. The inferior mesenteric vein lies under the parietal peritoneum, directly behind the lateral end of this fold. The recess extends superiorly and varies in size, but commonly admits a fingertip.

Inferior duodenal recess

The inferior duodenal recess is usually present, often in association with a superior recess, with which it may share an orifice. It lies to the left of the fourth part of the duodenum, adjacent to the third lumbar vertebra, behind an avascular, triangular inferior duodenal fold, which has a sharp upper edge. It usually admits one or two fingertips and extends inferiorly, sometimes behind the fourth part of the duodenum and to the left, in front of the ascending branch of the left colic artery and the inferior mesenteric vein.

Paraduodenal recess

The paraduodenal recess may occur in conjunction with superior and inferior duodenal recesses. It lies a little to the left and slightly behind the fourth part of the duodenum, behind a falciform fold of peritoneum. The free right edge of this fold contains the inferior mesenteric vein and ascending branch of the left colic artery, and is part of the left colic mesentery. This recess is the site of a 'left paraduodenal hernia' (Khan et al 1998).

Retroduodenal recess

The retroduodenal recess is the largest of the duodenal recesses but is rarely present. It lies behind the third and fourth parts of the duodenum in front of the abdominal aorta, is up to 10 cm deep, and has a wide orifice flanked by duodenoparietal peritoneal folds.

The distribution of peritoneal metastases via the paracolic gutters is not the same on each side. Peritoneal fluid usually tracks freely along the right paracolic gutter, whereas the flow of peritoneal fluid running down the left paracolic gutter from the upper abdomen will be impeded by any attachments of the sigmoid colon to the left abdominal wall. Cancer deposits at this site tend to invade the mesentery of the junction between the sigmoid and descending colon, and therefore necessitate sigmoid colectomy for tumour clearance.

The superior and inferior recesses of the duodenum may be sites of occult peritoneal metastases and must be examined in patients having surgery for peritoneal metastases.

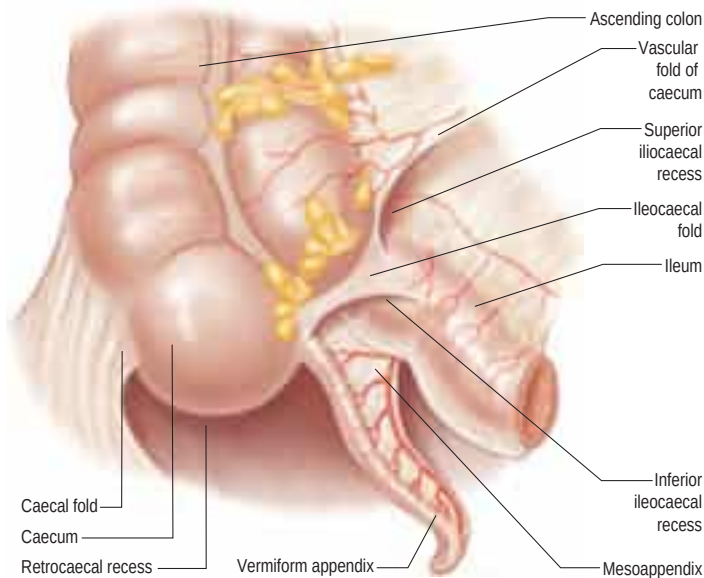


Fig. 63.10 The peritoneal folds and recesses in the caecal region.

Duodenojejunal recess

The duodenojejunal or mesocolic recess occurs in a minority of adults. When present, it is almost never associated with other duodenal recesses. It is approximately 3 cm deep and lies to the left of the abdominal aorta, between the duodenojejunal junction and the root of the transverse mesocolon. It is bounded above by the pancreas, on the left by the kidney, and below by the left renal vein. Its circular opening lies between two peritoneal folds and faces down and to the right.

Mesentericoparietal recess

The mesentericoparietal recess or fossa is rarely present. It lies just below the third part of the duodenum and invaginates the upper part of the mesentery of the small intestine towards the right. Its orifice is large and faces to the left, behind a fold of mesentery raised by the superior mesenteric artery. This recess is the site of the rare 'right paraduodenal hernia'.

Caecal recesses

Peritoneal folds around the caecum may form a variety of recesses (**Fig. 63.10**); these have the potential to become sites of internal herniation (**Rivkind et al 1986**). Paracaecal recesses are common sites for abscess formation after acute appendicitis.

Superior ileocaecal recess

The superior ileocaecal recess is usually best developed in children and may be absent in the aged or obese. It is a narrow slit bounded in front by a vascular fold of peritoneum containing the anterior caecal vessels, behind by the ileal mesentery, below by the terminal ileum and on the right by the ileocaecal junction. Its orifice opens to the left.

Inferior ileocaecal recess

The inferior ileocaecal recess is similarly well developed in the young but frequently obliterated by fat in adults. It is formed by the ileocaecal fold, which extends from the anteroinferior aspect of the terminal ileum to the front of the mesoappendix (or to the appendix or caecum). It is also known as the 'bloodless fold of Treves', although it sometimes contains blood vessels that bleed if divided during surgery. If inflamed, especially when the appendix and its mesentery are retrocaecal, it may be mistaken for the mesoappendix. The recess is bounded in front by the ileocaecal fold, above by the terminal ileum and its mesentery, to the right by the caecum, and behind by the upper part of the mesoappendix. Its orifice opens downwards to the left.

Retrocaecal recess

The retrocaecal recess lies behind the caecum. It ascends behind the ascending colon to a variable extent, often being large enough to admit an entire finger. It is bounded in front by the caecum (and sometimes the proximal ascending colon), behind by the parietal peritoneum, and on each side by caecal folds (parietocolic folds) passing from the caecum to the posterior abdominal wall. The vermiform appendix frequently occupies this retrocaecal recess (**p. 1142**).

In addition to these caecal recesses, one or more shallow paracaecal or paracolic recesses may be present. These are simply depressions within the right paracolic gutter and are not sites of internal herniation.

Intersigmoid recess

The intersigmoid recess is present in fetal life and infancy, but often disappears subsequently. It lies behind and inferior to the apex of the V-shaped attachment of the sigmoid mesocolon. It opens downwards and varies in shape from a slight depression to a shallow fossa. Its posterior wall is formed by the parietal peritoneum of the posterior abdominal wall, where it covers the left ureter as it crosses the bifurcation of the left common iliac artery. It is a rare site for an internal hernia (**Harrison et al 2011**).

Internal hernias

An internal hernia is a protrusion of a viscus through a normal or abnormal aperture (which includes a surgically created opening) within the confines of the peritoneal cavity (**Martin et al 2006**). It is possible for any of the peritoneal recesses, foramina or cul-de-sacs described in this chapter to be associated with the development of an internal hernia, but with a clinically apparent internal hernia the relevant peritoneal recess is usually enlarged and its aperture narrowed compared to normal. Among the many potential sites for an internal hernia, the most common are paraduodenal, followed by pericaecal, and hernias through the epiploic foramen.

ADDITIONAL CLINICAL ASPECTS OF THE PERITONEAL CAVITY

Peritoneal fluid collections

Fluid collections frequently develop within the peritoneal cavity in response to a wide range of pathological processes. In the absence of inflammation, peritoneal adhesions or previous surgery, serous fluid tends to be distributed widely between the peritoneal spaces. Simple ascites can therefore be drained freely from any convenient dependent part of the peritoneal cavity; drainage is most commonly performed by blind puncture or ultrasound-guided insertion of a catheter into the lower left or right paracolic gutters. The mobility of the small bowel makes it very unlikely to be injured during this procedure.

Fluid collections caused by inflammatory processes are often much thicker because they contain pus, fibrin or blood. Furthermore, they are usually associated with peritoneal adhesions arising from peritoneal inflammation. These factors predispose to the formation of localized fluid collections, which may become walled-off as the inflammatory process progresses. Any of the peritoneal spaces may develop a localized collection, but the subphrenic, subhepatic and pelvic spaces are the most common sites since they are well defined by peritoneal folds and organs forming their boundaries. These spaces are also the most dependent regions within the peritoneal cavity.

Surgical access to the peritoneal spaces is less often needed today because of advances in radiologically guided percutaneous drainage using fluoroscopy, ultrasound, computed tomography (CT) or magnetic resonance imaging (MRI) guidance. These techniques offer reliable and versatile methods of accessing peritoneal spaces, including relatively inaccessible subhepatic, perihepatic or intermesenteric collections. Posterolateral translumbar or trans-sciatic approaches can also be used to access the retroperitoneum and pelvis, respectively. Occasionally, a surgical approach is necessary; thus, a subcostal or lateral intercostal incision may be required to drain a subphrenic abscess or an inguinal incision to drain a pelvic abscess.

Peritoneal dialysis

The mean surface area of the peritoneum in the adult (female) has been estimated to be about 1.4 m² (with the visceral peritoneum accounting for about 80% of the total) (**Albanese et al 2009**). Mesothelium resembles vascular endothelium in that it allows the passage of ions and small molecules. Normally, the volume of fluid transmitted by peritoneal surfaces is small, but large volumes of fluid can be instilled into the peritoneal cavity and then siphoned out, using the peritoneum as a dialysing membrane. This can be used to support individuals with acute or chronic renal failure.

Ventriculoperitoneal shunts

The absorptive capabilities of the peritoneum can be exploited to absorb excess transitional fluids from several sites in the body. The most common of these is the absorption of cerebrospinal fluid diverted into the peritoneal cavity using a shunt (a fine catheter) from the cerebral ventricles or the intrathecal space. The catheter placed within the peritoneal cavity can be equipped with a one-way valve to prevent reflux of peritoneal fluid into the cerebrospinal fluid. The cerebrospinal fluid is continuously absorbed by the peritoneum, maintaining a low pressure within the intraventricular or intrathecal space.

Intraperitoneal cancer chemotherapy

In patients with peritoneal metastases, attempts to optimize the benefits of cytoreductive surgery may involve the administration of

chemotherapeutic agents (diluted in large volumes of fluid) directly into the peritoneal cavity. Peritoneal access may be achieved via an intraperitoneal catheter connected to a subcutaneous port or by repeated paracentesis. The infused chemotherapeutic agents may be retained locally for longer than systemically administered agents. Targeting drugs to the peritoneal space may improve the results of treatment if complete tumour resection is not possible.



Bonus e-book video

Video 63.1 Surgical exploration of the peritoneal cavity.

KEY REFERENCES

- Coakley FV, Hricak H 1999 Imaging of peritoneal and mesenteric disease: key concepts for the clinical radiologist. *Clin Radiol* 54:563–74.
An explanation of the complex anatomy of the upper abdominal peritoneal fold suspending the stomach, liver and spleen.
- Healy JC, Reznick RH 1999 The anterior abdominal wall and peritoneum. In: Butler P, Mitchell A, Ellis H (eds) *Applied Radiological Anatomy*. Cambridge: Cambridge University Press, pp. 189–200.
A demonstration of the imaging anatomy of the peritoneal spaces and reflections using cross-sectional imaging.
- Martin LC, Merkle EM, Thompson WM 2006 Review of internal hernias: radiographic and clinical findings. *AJR* 186:703–17.
- Meyers M 1994 *Dynamic Radiology of the Abdomen: Normal and Pathologic Anatomy*. New York: Springer.
- Meyers MA, Oliphant M, Berne AS et al 1987 The peritoneal ligaments and mesenteries: pathways of intraabdominal spread of disease. *Radiology* 163:593–604.
A systematic application of anatomic and dynamic principles to the understanding and diagnosis of intra-abdominal disease.
- Pestieu SR, Wolk R, Sugarbaker PH 2000 Congenital pleuroperitoneal communication in a patient with pseudomyxoma peritonei. *J Surg Oncol* 73:174–8.
- Platell C, Cooper D, Papadimitriou JM et al 2000 The omentum. *World J Gastroenterol* 6:169–76.
A comprehensive review of the greater omentum.
- Schnüriger B, Barmparas G, Branco BC et al 2011 Prevention of postoperative peritoneal adhesions: a review of the literature. *Am J Surg* 201:111–21.
- Simmons LE, Mir AR 1989 A review of management of pleuroperitoneal communication in five CAPD patients. *Adv Perit Dial* 5:81–3.
- Stringer MD 2009 *Eponyms in Surgery and Anatomy of the Liver, Bile Ducts and Pancreas*. London: Royal Society of Medicine Press.
A short reference book that contains information and illustrations relating to the original descriptions and the authors of many of the eponyms cited in this chapter (e.g. Cullen, Winslow, Morison).

REFERENCES

- Albanese AM, Albanese EF, Mino JH et al 2009 Peritoneal surface area: measurements of 40 structures covered by peritoneum: correlation between total peritoneal surface area and the surface calculated by formulas. *Surg Radiol Anat* 31:369–77.
- Ashaolu JO, Ukwenya VO, Adenowo TK 2011 Cystoduodenal ligament as an abnormal fold and the accompanying anatomical and clinical implications. *Surg Radiol Anat* 33:171–4.
- Autio V 1964 The spread of intraperitoneal infection. *Acta Chir Scand Suppl* 36:5–31.
- Baessler K, Schuessler B 2000 The depth of the pouch of Douglas in nulliparous and parous women without genital prolapse and in patients with genital prolapse. *Am J Obstet Gynecol* 182:540–4.
- Boaz NT, Martin AH, Thompson K et al 2011 Structure and functional significance of the transverse vesical fold. *Clin Anat* 24:62–9.
- Coakley FV, Hricak H 1999 Imaging of peritoneal and mesenteric disease: key concepts for the clinical radiologist. *Clin Radiol* 54:563–74.
An explanation of the complex anatomy of the upper abdominal peritoneal fold suspending the stomach, liver and spleen.
- Coulier B 2009 64-row MDCT review of anatomic features and variations of the normal greater omentum. *Surg Radiol Anat* 31:489–500.
- Feldberg MA, van Leeuwen MS 1990 The properitoneal fat pad associated with the falciform ligament. Imaging of extent and clinical relevance. *Surg Radiol Anat* 12:193–202.
- Hagiwara A, Takahashi T, Sawai T et al 1994 Milky spots as the implantation site for malignant cells in peritoneal dissemination in mice. *Cancer Res* 54:687–92.
- Harrison OJ, Sharma RD, Niayesh MH 2011 Early intervention in intersigmoid hernia may prevent bowel resection – a case report. *Int J Surg Case Rep* 2:282–4.
- Healy JC, Reznick RH 1999 The anterior abdominal wall and peritoneum. In: Butler P, Mitchell A, Ellis H (eds) *Applied Radiological Anatomy*. Cambridge: Cambridge University Press, pp. 189–200.
A demonstration of the imaging anatomy of the peritoneal spaces and reflections using cross-sectional imaging.
- Khan MA, Lo AY, Vande Maele DM 1998 Paraduodenal hernia. *Am Surg* 64:1218–22.
- Koninckx PR, Renaer M, Brosens IA 1980 Origin of peritoneal fluid in women: an ovarian exudation product. *Br J Obstet Gynaecol* 87:177–83.
- Martin LC, Merkle EM, Thompson WM 2006 Review of internal hernias: radiographic and clinical findings. *AJR* 186:703–17.
- Meyers MA 1973 Distribution of intra-abdominal malignant seeding: dependency on dynamics of flow of ascitic fluid. *Am J Roentgenol Radium Ther Nucl Med* 119:198–206.
- Meyers M 1994 *Dynamic Radiology of the Abdomen: Normal and Pathologic Anatomy*. New York: Springer.
- Meyers MA, Oliphant M, Berne AS et al 1987 The peritoneal ligaments and mesenteries: pathways of intraabdominal spread of disease. *Radiology* 163:593–604.
A systematic application of anatomic and dynamic principles to the understanding and diagnosis of intra-abdominal disease.
- Pestieu SR, Wolk R, Sugarbaker PH 2000 Congenital pleuroperitoneal communication in a patient with pseudomyxoma peritonei. *J Surg Oncol* 73:174–8.
- Platell C, Cooper D, Papadimitriou JM et al 2000 The omentum. *World J Gastroenterol* 6:169–76.
A comprehensive review of the greater omentum.
- Rivkind AI, Shiloni E, Muggia-Sullam M et al 1986 Paracecal hernia: a cause of intestinal obstruction. *Dis Colon Rectum* 29:752–4.
- Sanchez NC, Tenofsky PL, Dort JM et al 2001 What is normal intra-abdominal pressure? *Am Surg* 67:243–8.
- Schnüriger B, Barmparas G, Branco BC et al 2011 Prevention of postoperative peritoneal adhesions: a review of the literature. *Am J Surg* 201:111–21.
- Shimotsuma M, Shields JW, Simpson-Morgan MW et al 1993 Morphophysiological function and role of omental milky spots as omentum-associated lymphoid tissue (OALT) in the peritoneal cavity. *Lymphology* 26:90–101.
- Shrestha BM, Brown PW, Wilkie ME et al 2010 The anatomy and pathology of the lesser sac: implications for peritoneal dialysis. *Perit Dial Int* 30:496–501.
- Simmons LE, Mir AR 1989 A review of management of pleuroperitoneal communication in five CAPD patients. *Adv Perit Dial* 5:81–3.
- Stringer MD 2009 *Eponyms in Surgery and Anatomy of the Liver, Bile Ducts and Pancreas*. London: Royal Society of Medicine Press.
A short reference book that contains information and illustrations relating to the original descriptions and the authors of many of the eponyms cited in this chapter (e.g. Cullen, Winslow, Morison).
- Sugarbaker PH 2012 An overview of peritonectomy, visceral resections, and perioperative chemotherapy for peritoneal surface malignancy. In: Sugarbaker PH (ed) *Cytoreductive Surgery and Perioperative Chemotherapy for Peritoneal Surface Malignancy*. Textbook and Video Atlas. Woodbury, CT: Cine-Med Publishing, pp. 1–30.
- Wassilev W, Wedel T, Michailova K et al 1998 A scanning electron microscopy study of peritoneal stomata in different peritoneal regions. *Ann Anat* 180:137–43.
- Yoshikawa T, Hayashi N, Maeda E et al 2013 Peritoneal fluid accumulation in healthy men and postmenopausal women: evaluation on pelvic MRI. *AJR* 200:1181–5.
- Zonca P, Maly T, Mole DJ et al 2008 Treitz's hernia. *Hernia* 2:531–4.